Henry Kong
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Education

- University of California, Berkeley (August 2025 - Current)
- Legacy Magnet Academy High School (August 2021 - May 2025)
    - 4.0 GPA
- Irvine Valley College (August 2021 - May 2025)
    - 4.0 GPA

Extracurricular Activities

- Taipei Tech Racing (Formula SAE) 2023 - Current:
    - Accepted as a high-school student (the program is highly competitive and consists of only college students)
    - Created a 3D model of the actual car for reference using SolidWorks
    - Worked with the circuitry team to develop the motherboard for the car
    - Helped to create and weld together the chassis
- VEX Robotics (Builder & Programmer) 2023 - Current:
    - Competed at the state level
    - Competed at four events
    - Assistant Judge for two competition events in 2023
- Academic Decathlon (15+ medals) 2021-2023:
    - Economics: 1st place
    - Art: 2nd place
    - Music: 4th place
    - Math: 4th place
    - Overall Team: 2nd place
- AWS DeepRacer (Student Coach) 2020 - 2023:
    - Orlando Regional Competition: 1 of 3 Judges (4 years of experience)
    - First middle schooler to be accepted into the program
    - Coached over 40 students on machine learning algorithms
- STEM Summer Research
    - Wrote a proposal with a professor from CalTech for an experiment to prove string theory
- President of Pickleball Club
- President of IVC Help Club
- Catch The Wave Passion Project

    - Recorded interviews with the entire school faculty to introduce them to new students at LMA

Internships:

- Systems Integrated - Paid Engineering Intern (7/1/2025 - 8/10/2025)
    - Built an offline ping monitor on an Ubuntu VM (ESXi) with zero external dependencies; scanned a /24 every 30s and logged IP/hostname/latency/timestamp to CSV (256 hosts).
    - Wrote Bash scanner with systemd service & timer; added on-demand "Run Scan Now" trigger.
    - Delivered a no-framework web GUI (BusyBox netcat + HTML/JS) on port 8080 with auto-refresh, sortable columns, and green/red status highlighting.
    - Implemented resilient hostname resolution (DNS PTR lookups with CSV cache fallback); tuned Netplan & systemd-resolved for correct nameserver.
    - Hardened reliability: fixed Exec/CRLF issues, sanitized CSV/HTML output, and removed stray newlines for stable rendering.
    - Operated 3D printers to produce prototypes and mounting components; applied Linux-based tools to configure automation hardware; collaborated with senior engineers to improve documentation and layouts.
    - Outcome: Zero-install, low-overhead monitoring with instant rescans and clear, sortable visibility into live reachability and response times.
- Systems Integrated - Engineering Intern (6/10/2024 - 7/12/2024)
    - Worked with the head of software engineering at the headquarters of Systems Integrated on the wiring diagram of a well pump that controlled water/water filtration for over 30,000 residents
    - Built a computer for work use from scratch
    - Calculated the parameters for the setup of radio dishes for the Suffolk County Water Authority
    - Designed and improved the wiring for a water plant in San Diego
    - Designed a well pump using AutoCAD
    - Collaborated with a fellow intern to create and wire physical circuits
- TPSF Summer Camp Teacher's Assistant at Orchard Hills School (Summers of 2022 & 2023)
    - Taught marine biology and physics to over 60 students
- [Paid] Clean & Green Technology (2021-2024)
    - Engineered and developed a prototype automotive battery monitoring and analysis system, which was later successfully commercialized and published for widespread industry use.
    - $100,000+ USD in monthly revenue globally (1,000+ units at ~$100 each)

Accomplishments

- 32nd National Environmental Protection and Energy-Saving Car Competition (Taipei Tech Racing)
    - First Place Best Energy-Saving Award
    - Fifth Place Dynamic Competition Award (Overall Performance)
- Received over TWD 50,000 (About $1,500) from scholarships and tournament prizes from FSAE
- Published Researcher at Curieux Academic Journal